ENPM 701 – Autonomous Robotics: Spring 2026

Master of Engineering Program in Robotics

Due Date Friday, March 27th, 2026

Submission Information

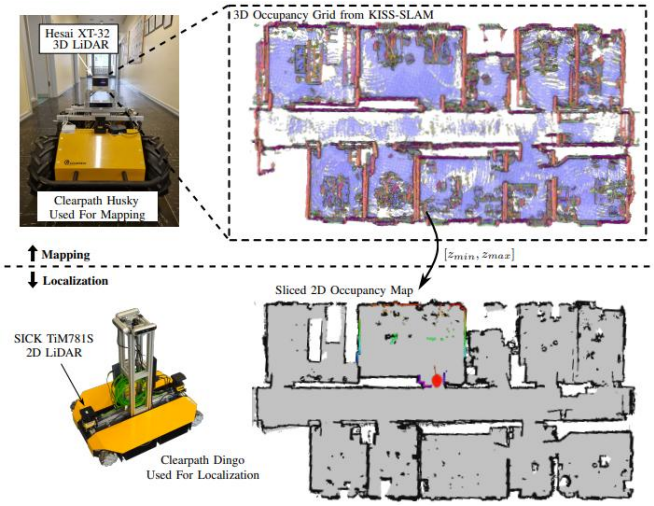
- This assignment focuses on completing assembly of the course robotic ground vehicle, in conjunction with an initial exploration of localization
- Submit responses to required Questions via Gradescope by 11:59 pm

Reading: In preparation for our Localization module, please read through Chapter 5 of the Seigwart text "Autonomous Mobile Robots," available on ELMS.

On a related note, Guadagnino et. al. have put together a great paper on the use of Lidar for SLAM, entitled "KISS-SLAM: A Simple, Robust, and Accurate 3D LiDAR, SLAM System With Enhanced Generalization Capabilities." The paper and the associated GitHub are available via the links below. Please read through the paper at your convenience:

KISS-SLAM paper

KISS-SLAM GitHub



Question #1 (0 points)

A general theme of my teaching philosophy is that I am here for you, the student, and continually strive to add value to your academic experience. In this spirit, please log onto ELMS and complete the "Mid-Session Feedback Survey" located under ELMS > Quizzes. *All feedback is* ***anonymous*** (I have no indication who submits what information) and is gathered with the intent of improving the quality of ENPM 701 such that you get the maximum benefit out of the course. **Please complete by 11:59 pm Friday March 27th**.

Question #0.1 (nothing to submit)

General reminder to use your phone to **record images and 30-60 second video clips throughout the course**. These images/videos can then be stitched together towards the end of the Spring as part of your video documentary submission for the Grand Challenge. It is always better to have to cut out extra material than to say "I wish I had an image/video of that…!" at the conclusion of the project. And, as always…***have fun with the filming process***!

Looking for inspiration for your documentary video? There are a host of great video submissions from prior ENME 701 students on the course YouTube channel!

ENPM 701 YouTube Playlist

Question #1 (5 points)

On the topic of robotic perception, Mark Rober has put together an interesting video exploring the limitations of passive sensor architectures. In 3-4 paragraphs, summarize the results and impacts of the material presented in the following video:

YouTube: Can You Fool a Self Driving Car?

In particular, be sure to think about and discuss potential edge cases for automobiles using camera-only perception architectures, and how inclusion of additional sensor types (passive and/or active) might improve the vehicle's performance within the confines of those edge cases.

Question #2 (20 points)

The primary focus of this assignment is to complete the assembly of your ground vehicle. With this in mind, head to ELMS > Modules > 05 and **download the notes** for this week's lecture.

A time-lapse video of the full assembly process is available on the course YouTube page:

Pirate Assembly Video (Pirate)

Baron Assembly Video (Baron)

Reminder that it is imperative to ***take your time*** and methodically assemble the ground vehicle. Rushing through the integration process now will almost certainly lead to an exponential increase in debugging time down the road.

To complete this portion of the assignment:

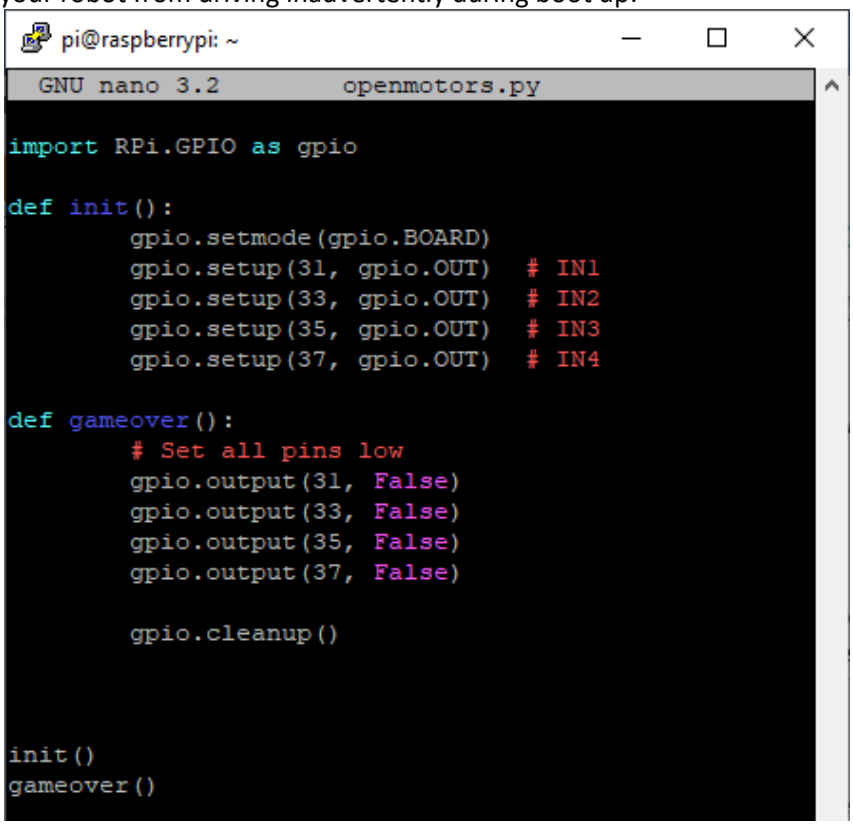
1. Complete the mechanical and electrical assembly of the ground vehicle as detailed in the lecture notes.

2. Copy/paste your response to the first In-Class Exercise ("Using the circuit below…") into the .pdf uploaded to Gradescope. This Exercise asks you to sketch the time-domain control signals for an H-bridge. Be sure to include a few sentences detailing (a) what is an H-bridge and how does it work?, and (b) justifying your answers for the time-domain signals.

3. Create and test the *motorcontrol01.py* and *motorcontrol02.py* scripts as detailed in the lecture notes. This should include completion of the second In-Class Exercise ("Update the…"), although there is nothing to submit for this second In-Class Exercise.

4. To help prevent your robot from driving away during a power cycle, create the following *openmotors.py script* and ***run the script each time you boot up your Pi***. Details on how to run a script on bootup are available here: Run Program at Startup

   The *openmotors.py* script sets all GPIO pins tied to the H-bridge as false (i.e. GND), helping to prevent your robot from driving inadvertently during boot up:




```
import RPi.GPIO as gpio

def init():
        gpio.setmode(gpio.BOARD)
        gpio.setup(31, gpio.OUT)  # IN1
        gpio.setup(33, gpio.OUT)  # IN2
        gpio.setup(35, gpio.OUT)  # IN3
        gpio.setup(37, gpio.OUT)  # IN4

def gameover():
        # Set all pins low
        gpio.output(31, False)
        gpio.output(33, False)
        gpio.output(35, False)
        gpio.output(37, False)

        gpio.cleanup()



init()
gameover()
```

5. Using your cell phone camera or equivalent, record a minimum 30 second video clip of yourself teleoperating the ground vehicle with the *motorcontrol02.py* script. This should include clear demonstration of the forward( ), reverse( ), pivotleft( ), and pivotright( ) functions. **Upload the video to your YouTube account** then include a link to the video in the .pdf uploaded to Gradescope.

Question #3 (5 points)

This semester's Grand Challenge focuses on autonomy in AgTech. One company working in this domain is Burro:

Burro

Burro builds collaborative autonomous robots (affectionately known as "Burros") that work alongside human crews in farms, nurseries, and vineyards. These robots automate tasks like hauling and towing, enabling workers to focus on higher-value activities.

Based on the information presented on the Burro website, as well as your own research, discuss in 2-3 paragraphs the Localization architectures the Burro platform uses.